

Singari Sai Ashwatha

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Education

Manipal Institute of Technology, Manipal, India | *July 2023 – July 2027*

- B.Tech., Computer and Communication Engineering — **CGPA: 8.77 / 10**
- **Courses:** Data Structures and Algorithms, Object Oriented Programming, Database Management Systems, Computer Networks, Embedded Systems, Operating Systems

Technical Skills

- **Languages:** C, C++, Python, Java, JavaScript, HTML, CSS, SQL
- **Frameworks:** React.js, Node.js, Express.js, Tailwind CSS, LangChain, Socket.IO
- **Tools & Fundamentals:** LabVIEW, MongoDB, MySQL, REST APIs, API Integration, JWT Authentication, Git, Postman, Generative AI, Jupyter Notebook

Experience

Software Engineering Intern, NI (National Instruments), now part of Emerson | *Jul 2026 – Jun 2027*

- Developing **high-performance C++ APIs and hardware drivers** for **RF Signal Acquisition (RFSA)** and **RF Signal Generation (RFSG)** instruments, enabling compatibility with **LabVIEW**, NI software, and third-party applications.
- Building and maintaining production-grade software by implementing new features, debugging complex issues, validating hardware-software interactions, and optimizing reliability across RF measurement systems.
- Collaborating with cross-functional RF R&D engineers to design, test, and deliver robust engineering solutions while following industry-standard software development and version control practices.

Projects

Sentinel - Encrypted Traffic Threat-Intelligence Engine (C++17, Python, Node.js, Docker)

- Developed a multithreaded C++17 deep packet inspection engine with a lock-free worker pool to parse Ethernet/IP/TCP/UDP traffic, achieving throughput of **~247K packets/sec**.
- Implemented **JA3 TLS fingerprinting** and built real-time threat detectors for port scanning, DNS tunnelling, C2 beaconing, SYN flood attacks, and cleartext credential exposure using flow-based analysis.
- Built a **Random Forest-based encrypted traffic classifier** using flow statistics and integrated structured telemetry with a real-time WebSocket dashboard, supported by Docker, GitHub Actions CI, and unit tests.

CVision - AI-powered Resume Builder (MERN Stack)

- Engineered a high-performance editor featuring real-time side-by-side previewing and section-wise editing across multiple ATS-optimized themes.
- Integrated LangChain for AI-driven content generation and ImageKit for automated background removal, reducing resume creation time.
- Developed automated parsing for instant data import and implemented high-fidelity PDF export to ensure consistent formatting retention.

Certificates

- **Node.js & MongoDB: Developing Back-end Database Applications**
- **JavaScript Programming Essentials**

[Certificate](#)
[Certificate](#)

Clubs and Activities

Volt-a-Bit, Manipal Institute of Technology

- Contributing to the development of a 32-bit operating system using **C** and **x86 Assembly**.
- Working on bootloader development and emulator setup.